Identification and inference for the shape of a distribution from rounded reports

Samir Orujov*

Abstract

Survey incomes are reported in round numbers, and tests of unimodality applied to such reports reject for every unimodal income distribution we simulate. We study what the reports identify. Conditional on the rounding level ℓ, the greatest common divisor G of a report with the least common multiple W of the candidate levels satisfies $\mathbb{P}(G = d \mid \ell) = \ell\,\varphi(W/d)/W$, with φ Euler's totient. The rounding probabilities are therefore identified under a regularity condition on the residues of income modulo W, and the residue of a report modulo W is uniformly distributed given its divisibility pattern, which yields a specification test free of nuisance parameters. At any width that is a common multiple of the active levels the cell probabilities of latent income are identified up to a second-order boundary term, and binning preserves unimodality; we test unimodality of the cell vector with a moment-selection bootstrap whose asymptotic level is at most nominal at every point of the null, including its boundary. Finer resolution is identified by inverting the reporting map, but the inversion needs the rounding profile conditional on latent income, which the reports identify only coarsely. On American data the specification test rejects the mechanism for every earnings variable, and a thinning in the disposable income of one-person households turns out to be rounding in gross earnings carried through the tax computation.

*School of Business, ADA University, Baku, Azerbaijan, and Information and Communication Technologies Agency of Azerbaijan. sorujov@ada.edu.az.

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1 Introduction

Among Californians aged 25 to 64 in the 2022 American Community Survey (ACS), 88% of positive wage and salary reports are multiples of \$1,000, 55% of \$5,000 and 38% of \$10,000; in the Current Population Survey (CPS) the shares are 86%, 54% and 35%, and a further 15% of reports are multiples of \$25,000. Such heaping perturbs a mean very little, but a kernel estimate of reported earnings has a local maximum at every round number, and a test of unimodality applied to the reports tests the questionnaire. In the simulations of Section 6, the excess mass test of Ameijeiras-Alonso et al. (2019) and the dip test of Cheng and Hall (1998) reject unimodality in every one of 16,800 replications drawn from seven unimodal latent densities and reported through three rounding mechanisms, and hold their level with unrounded data wherever the latent density is unimodal on the scale on which they are applied.

Bimodality, polarization and a thinning middle of the income distribution are established topics in the study of inequality (Esteban and Ray, 1994; Wolfson, 1994; Jenkins, 1995; Quah, 1996; Cowell et al., 1996; Duclos et al., 2004; Pittau et al., 2010), and all rest on the shape of a density estimated from such reports. This paper asks what features of the shape of a latent income distribution are identified from rounded reports, and how inference about them should be calibrated.

Model and contributions. Let X be latent income and let the report be $Y = r_L(X)$, where r_ℓ rounds to the nearest multiple of ℓ and the level L is drawn from a finite ladder $\mathcal{L} \ni 1$ with probabilities $\pi_\ell(x) = \mathbb{P}(L = \ell \mid X = x)$ that may depend on income. Write $W = \operatorname{lcm} \mathcal{L}$.

Identification of the mechanism. Conditionally on $L = \ell$ the report is ℓ times an integer, and all the arithmetic information it carries about the level is contained in $G = \gcd(Y, W)$. If the residue of that integer modulo W/ℓ is uniform, a condition on the income distribution that we bound through its characteristic function, then $\mathbb{P}(G = d \mid L = \ell) = \ell\,\varphi(W/d)/W$ for $\ell \mid d \mid W$, with φ Euler's totient, and the marginal rounding probabilities are identified from the divisibility pattern of the reports (Proposition 1). Conditionally on its pattern, the residue of a report modulo W is uniform over the residues with that pattern, whatever the mechanism, which yields a specification test with an exact conditional calibration and no nuisance parameter (Theorem 1), sensitive to the failure that matters: reports crowding into part of the period $[0, W)$ because income does not span many periods of W.

Resolution. Rounding to a level that divides the cell width moves mass only across cell boundaries, and if boundary reports are split between the two cells the reported cell probabilities equal the latent ones up to a term of second order in the level (Lemma 2). At the least common multiple w_0 of the active levels the latent cell vector is therefore identified without any model of how respondents choose a level, and since binning preserves unimodality (Lemma 3), unimodality of the latent density becomes the hypothesis that a multinomial probability vector lies in the cone of unimodal sequences.

Inference at that resolution. We measure departure from the cone by the discrete excess mass and calibrate it by resampling from a projection onto the cone after adjacent cells whose difference is within κ_n standard errors have been pooled, as in generalized moment selection (Andrews and Soares, 2010). This bootstrap reproduces the limit law of the statistic at every fixed point of the null, including points on its boundary, where resampling from the unmodified projection does not (Theorem 2).

Finer resolution, and its price. Below w_0 the reported cell vector is a known linear map of the latent one once the profile $\pi_\ell(\cdot)$ is known, and for ladders totally ordered by divisibility the map is invertible at every width down to the finest active level (Theorem 3). The profile needed is conditional on latent income, not on the report; band fits of the mixture identify

its band averages, but the inversion amplifies small errors in any estimated profile and in our simulations holds its level at large sample sizes only when the profile is known, so we present it as a way to use a profile measured outside the survey, gated by the specification test.

Evidence. On every American earnings variable we examine the specification test rejects the mechanism, for an identifiable reason: the ladder's least common multiple is \$50,000 in most samples and between two fifths and two thirds of reports lie below it, so the residues cannot be close to uniform, and in the ACS the Census Bureau's own rounding of released amounts violates the condition by construction. The certified width then leaves four to ten cells: the reports support no statement about the modality of American earnings, and the procedure says so rather than returning a rejection that reflects the questionnaire. On monthly housing costs the ladders are fine; a valley at the bottom of the distribution of gross rent does not survive a coarser grid and is not retained. On disposable income of one-person households in the CPS the lattice enters with transfer-dependent households, and once they are removed the specification test accepts the mechanism in four of five years. A thinning between \$25,000 and \$32,000 is significant only among respondents whose gross earnings are multiples of \$5,000: it is rounding in gross earnings carried through the tax computation to amounts that are no longer round, which a specification test applied to the constructed variable cannot see; the test belongs on the amount that was reported.

Related literature. Demography summarizes digit preference in reported ages by indices of the share of reports ending in zero or five (Whipple, 1919; Myers, 1940; A'Hearn et al., 2009). Heitjan and Rubin (1990, 1991) place coarsened data in the missing-data framework; a mechanism in which the level depends on income is not coarsening at random and must be estimated. Wright and Bray (2003), Wang and Heitjan (2008) and Bar and Lillard (2012) fit rounding as a mixture over levels jointly with a parametric model for the latent variable; Camarda et al. (2008) estimate digit preference jointly with a penalized latent

density in the composite link model of Eilers (2007); Groß and Rendtel (2016), Zinn and Würbach (2016) and Drechsler and Kiesl (2016) treat heaped income by stochastic EM or multiple imputation, again jointly with a model of the density. These joint approaches replace our residue condition by smoothness of the latent density at the scale of the finest level; neither assumption dominates, and Section 7 shows data in which ours fails and is detected. The closest antecedents of our identification argument are Ball et al. (1985) and Harris (1991), who estimate the probabilities with which prices are quoted to successively coarser resolutions from the clustering of observed prices; we make the divisibility structure explicit, its law being a matrix of totients whose triangular structure, with the positive definiteness of greatest-common-divisor matrices (Smith, 1875; Beslin and Ligh, 1989), gives identification, and whose ancillary part supplies a specification test. The Fourier analysis of rounding goes back to Sheppard (1898); Janson (2006) treats it in generality, and Heitjan (1989) reviews inference from grouped data. In economics, rounding has been read as a signal of uncertainty (Manski and Molinari, 2010; Kleinjans and van Soest, 2014; Ruud et al., 2014; Binder, 2017), the interval a report represents being the object of interest.

The modality literature (Silverman, 1981; Hartigan and Hartigan, 1985; Müller and Sawitzki, 1991; Cheng and Hall, 1998; Hall and York, 2001; Fisher and Marron, 2001; Dümbgen and Walther, 2008; Ameijeiras-Alonso et al., 2019) is built for continuous data. At the certified width our problem is order-restricted inference for a multinomial (Robertson et al., 1988; Silvapulle and Sen, 2005); maximum likelihood estimation of a unimodal probability mass function is studied by Balabdaoui and Jankowski (2016). The failure of the bootstrap on the boundary of a parameter space is due to Andrews (2000), and our calibration follows the moment-selection approach of Andrews and Soares (2010) and the directional-differentiability framework of Fang and Santos (2019). Partial identification from interval-valued reports is developed by Manski and Tamer (2002).

Proofs not given in the text, and additional tables, are in the supplementary material.

2 The reporting mechanism

2.1 Model

Let $X > 0$ denote latent income, with density g; amounts are measured in the smallest currency unit, so reports are integers. A candidate ladder is a finite set of positive integers $\mathcal{L} = \{\ell_1 < \cdots < \ell_J\}$ with $\ell_1 = 1$, and $W = \operatorname{lcm} \mathcal{L}$. A respondent with income x chooses a level $L \in \mathcal{L}$ with probability $\pi_\ell(x) = \mathbb{P}(L = \ell \mid X = x)$ and reports

$$Y = r_L(X), \qquad r_\ell(x) = \ell K_\ell, \qquad K_\ell = \lfloor x/\ell + \tfrac{1}{2} \rfloor.$$

Write $\bar{\pi}_\ell = \mathbb{E}\, \pi_\ell(X)$ for the marginal rounding probabilities. Nothing is assumed about the dependence of $\pi_\ell(\cdot)$ on income.

Since every $\ell \in \mathcal{L}$ divides W, the divisibility pattern of a report,

$$S(Y) = \{\ell \in \mathcal{L} : \ \ell \mid Y\},$$

depends on Y only through $G = \gcd(Y, W)$, the *arithmetic content* of the report. For $d \mid W$ write $S(d) = \{\ell \in \mathcal{L} : \ell \mid d\}$, and let $\tau(W)$ be the number of divisors of W and V the number of distinct sets $S(d)$, $d \mid W$; these are the attainable patterns.

2.2 Identification

Assumption 1 (Residue regularity)**.** For every $\ell \in \mathcal{L}$, conditionally on $L = \ell$, the residue of K_ℓ modulo W/ℓ is uniformly distributed on $\{0, 1, \ldots, W/\ell - 1\}$.

Assumption 1 holds exactly for no income distribution and approximately for any whose relevant densities vary slowly on the scale W; the next lemma, a form of the Poisson-summation argument behind Sheppard's corrections (see Janson, 2006), makes this quantitative.

Lemma 1. *Fix $\ell \in \mathcal{L}$, put $m = W/\ell$, and let ϕ_ℓ be the characteristic function of X given $L = \ell$. Suppose the W-periodization of the conditional density of X given $L = \ell$ is square integrable on $[0, W)$. Then for every residue r,*

$$\Big|\mathbb{P}\big(K_\ell \equiv r \pmod m \mid L = \ell\big) - \frac{1}{m}\Big| \;\le\; 2\sum_{t\ge 1} \min\Big(\frac{1}{m}, \frac{1}{\pi t}\Big)\big|\phi_\ell(2\pi t/W)\big|.$$

The bound is governed by the decay of ϕ_ℓ at the frequencies $2\pi t/W$, that is, by how many periods of W the conditional distribution of income spans. For a lognormal calibrated to American earnings (median \$40,135, interquartile range \$21,816 to \$73,574) the leading term $|\phi(2\pi/W)|$ is 0.005 when $W = 10{,}000$ and 0.139 when $W = 50{,}000$. An indicator available before any estimation is the share of reports below W: when most reports lie in the first period of W, the residues cannot be close to uniform.

Proposition 1. *Under Assumption 1, for every $d \mid W$,*

$$\mathbb{P}(G = d \mid L = \ell) = \frac{\ell\,\varphi(W/d)}{W}\,\mathbf{1}\{\ell \mid d\}, \qquad \mathbb{P}(G = d) = \frac{\varphi(W/d)}{W}\sum_{\ell \in S(d)} \ell\,\bar\pi_\ell, \tag{1}$$

where φ is Euler's totient. Consequently:

(a) the $\tau(W) \times J$ matrix Λ with entries $\Lambda_{d\ell} = \ell\varphi(W/d)W^{-1}\mathbf{1}\{\ell \mid d\}$ has full column rank, and $\bar\pi$ is identified from the distribution of G;

(b) the pattern $S(G)$ is sufficient for $\bar\pi$;

(c) conditionally on $S(G) = S$, the arithmetic content has the law

$$\mathbb{P}\big(G = d \mid S(G) = S\big) = \frac{\varphi(W/d)}{\sum_{d' \in D_S}\varphi(W/d')}, \qquad d \in D_S = \{d \mid W : S(d) = S\}, \tag{2}$$

which does not depend on $\bar\pi$ or on the rounding profile $\pi_\ell(\cdot)$.

If Assumption 1 holds only up to $\varepsilon_\ell = \max_r |\mathbb{P}(K_\ell \equiv r \mid L = \ell) - \ell/W|$, then $|\mathbb{P}(G = d) - (\Lambda\bar\pi)_d| \le \varphi(W/d)\sum_\ell \bar\pi_\ell\varepsilon_\ell$.

Proof. On $\{L = \ell\}$, $G = \gcd(\ell K_\ell, W) = \ell \gcd(K_\ell, m)$ with $m = W/\ell$, so $\ell \mid G$. For $e \mid m$ the number of residues k modulo m with $\gcd(k, m) = e$ is $\varphi(m/e)$; with $e = d/\ell$ this is $\varphi(W/d)$, and uniformity of the residue gives the first identity in (1); averaging over L gives the second. For (a), order $\mathcal{L}$ increasingly and restrict Λ to the rows $d \in \mathcal{L}$. Since $\Lambda_{\ell'\ell} \neq 0$ requires $\ell \mid \ell'$, this $J \times J$ block is lower triangular, with diagonal entries $\ell\varphi(W/\ell)/W > 0$. Parts (b) and (c) follow from the factorization of (1) into a term depending on d only through $\varphi(W/d)$ and a term depending on d only through $S(d)$. The last statement follows by summing the error in the residue law over the $\varphi(W/d)$ residues with the required greatest common divisor. □

The rounding probabilities are estimated by maximum likelihood from the counts of patterns, which by (b) loses nothing, by the EM recursion for a finite mixture with known components. The arithmetic accounts for coincidences: once reports pile up at multiples of \$5,000, a fifth of them are multiples of \$25,000 whatever the respondents did, which a rule based on divisibility shares reads as a habit.

2.3 Testing the mechanism

The natural check, comparing observed divisibility shares with those implied by $\hat{\hat{\pi}}$, has almost no content on the ladders that occur in practice.

Proposition 2. *The attainable patterns are the sets $S(\operatorname{lcm} T)$, $T \subseteq \mathcal{L}$, and $V \geq J$. The model* (1) *imposes exactly $V - J$ restrictions on the distribution of the pattern. In particular $V = J$ if and only if $\mathcal{L}$ is closed under least common multiples, as it is when $\mathcal{L}$ is totally ordered by divisibility, and then the fitted pattern frequencies reproduce the observed ones exactly whenever the unconstrained solution lies in the simplex.*

For the ladder $\{1, 100, 1000, 5000, 10^4\}$ used in our simulations, and for the fifteen-level ladder used on American data in Section 7, $V = J$; for $\{1, 10, 100, 500, 1000, 2500, 5000, 10^4, 2.5 \cdot 10^4\}$, $V = J + 1$. A fit statistic built on patterns or on divisibility shares therefore cannot reject except at the boundary of the simplex, when the data ask for a negative rounding

probability. What reveals whether Assumption 1 holds is the position of the report within its arithmetic class. Let $R = Y \bmod W$. Under Assumption 1, $\mathbb{P}(R = r) = W^{-1} \sum_{\ell \in S(r)} \ell\, \bar{\pi}_\ell$ depends on r only through its pattern; hence

$$\mathbb{P}\big(R = r \mid S(R) = S\big) = \frac{1}{|C_S|}, \qquad r \in C_S = \{r \in \{0, \dots, W-1\} : \ S(r) = S\}, \tag{3}$$

whatever $\bar{\pi}$ and whatever the dependence of the level on income. Equation (3) contains Proposition 1(c), which groups residues by their greatest common divisor with W, and adds where in the period $[0, W)$ the reports of a class fall. When income does not span many periods of W, reports of every class crowd into the part of the period where income is concentrated, and (3) fails in a way no reweighting of the levels can absorb.

Theorem 1. *Let $Y_1, \dots, Y_n$ be independent copies of Y, and partition $[0, W)$ into M arcs of equal length. For each attainable pattern S and arc j let N_{Sj} be the number of reports with pattern S and residue in arc j, put $N_S = \sum_j N_{Sj}$, and let $p_{j|S} = |C_S \cap \mathrm{arc}_j|/|C_S|$. Define*

$$T_n = \sum_S \sum_j \frac{(N_{Sj} - N_S\, p_{j|S})^2}{N_S\, p_{j|S}},$$

the inner sum running over arcs with $p_{j|S} > 0$.

(a) Under Assumption 1, conditionally on $(N_S)_S$ the vectors $(N_{Sj})_j$ are independent multinomial with probabilities $p_{\cdot|S}$, whatever the mechanism. The conditional distribution of T_n is known, and T_n is asymptotically χ^2 with $\sum_S (M_S - 1)$ degrees of freedom, the sum running over patterns of positive probability and M_S being the number of arcs with $p_{j|S} > 0$.

(b) If for some pattern S with $\mathbb{P}(S(R) = S) > 0$ the conditional distribution of the arc of R given $S(R) = S$ differs from $p_{\cdot|S}$, then T_n/n converges almost surely to a positive constant, and the test that rejects for large T_n is consistent.

The test neither estimates $\bar{\pi}$ nor requires it to lie in the interior of the simplex, and

its p-value is computed exactly by conditional Monte Carlo. We use $M = 10$, exclude from the statistic and from its conditional distribution the classes with fewer than twenty reports, and with survey weights use weighted counts rescaled to the effective sample size $n_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$, a first-order Rao–Scott correction. We report T_n divided by its degrees of freedom, near one under the null, with the p-value.

The test rejects when the residues of the multipliers are not uniform, a property of the income distribution relative to W; when a level in use divides W but is absent from the ladder, since such reports sit at the wrong position within their class; and when reports carry mass at amounts that are not the result of rounding, if that mass is placed unevenly within a class; it cannot tell these apart. A rejection invalidates the estimated mixture, hence the active set, $\hat{w}_0$ and anything built on the profile, in particular the inversion of Section 5, but not Lemma 2, which holds at any common multiple of the levels in use whatever the mechanism; when the test rejects, a departure from unimodality found at $\hat{w}_0$ is therefore retained only if it persists at coarser widths that are multiples of $\hat{w}_0$ and of every candidate level plausibly in use. A level in use that does not divide W is invisible modulo W, so the candidate ladder must include every round amount respondents might use, in particular the coarse ones; levels of negligible estimated probability are then removed and the test rerun on the least common multiple of the active levels. An institutional amount that is a multiple of a level, a benefit of exactly \$9,000, say, is indistinguishable from rounding to \$1,000 unless it is large enough to distort the position of its class.

2.4 Active levels

Before fitting, candidate levels coarser than a quarter of the range between the first and 99th percentiles of the reports are removed, since the residues of their multipliers cannot be uniform; if the share of reports divisible by such a level exceeds the share implied by the fitted mixture by more than δ, the level is in use but not identified, and the procedure reports that no lattice-compatible grid exists. The active set is $\hat{A} = \{\ell \in \mathcal{L} : \hat{\tilde{\pi}}_\ell \geq \delta\}$ and the

certified width is $\hat{w}_0 = \operatorname{lcm} \hat{A}$, with $\hat{w}_0 = 1$ when $\hat{A}$ is empty. The cut-off is not a significance level: an ignored level costs a first-order error of order $\bar{\pi}_\ell$ in the cell probabilities (Section 3), detectable once it is large relative to the sampling error of order $n^{-1/2}$, so the cut-off must shrink with the sample size. Section 6 calibrates it on the least favourable designs, those with flat stretches, and we use $\delta_n = \min\{0.02,\, 2n_{\text{eff}}^{-1/2}\}$, which is 0.02 up to $n_{\text{eff}} = 10{,}000$ and 0.009 at 50,000. Since $\hat{\bar{\pi}} - \bar{\pi} = O_p(n^{-1/2})$ and δ_n is of the same order, a level with $\bar{\pi}_\ell = 0$ may be retained with non-vanishing probability, while every level with $\bar{\pi}_\ell > 0$ is retained with probability tending to one, because $\delta_n \to 0$. Hence $\hat{w}_0$ is a multiple of w_0, and therefore lattice-compatible, with probability tending to one: errors in the selection coarsen the grid without breaking it.

3 What the reports identify about the latent distribution

3.1 Cells at a lattice-compatible width

Let $A \subseteq \mathcal{L}$ be the set of levels in use. A width w is *lattice-compatible* if every $\ell \in A$ divides w, and the cells are $C_k = [kw - w/2,\, kw + w/2)$, $k \in \mathbb{Z}$. Write $P_k = \int_{C_k} g$ for the latent cell probabilities and $h_\ell = g\,\pi_\ell$, so that $g = \sum_\ell h_\ell$.

Rounding to a level that divides w keeps a report close to the latent amount, but it can still carry mass across a cell boundary. Whether it does depends on the parity of w/ℓ. If w/ℓ is odd, the boundary $b = kw + w/2$ is congruent to $\ell/2$ modulo ℓ, so it separates two rounding intervals of level ℓ and nothing crosses it. If w/ℓ is even, b is itself a multiple of ℓ, and every latent amount within $\ell/2$ of b is reported exactly at b. No choice of grid avoids this for all levels at once, because levels of different 2-adic order cannot all have odd ratio to the same width. The remedy is to split reports that fall exactly on a cell boundary equally between the two adjacent cells. Let q_k be the probability of cell C_k for the reports, computed

with this convention.

Lemma 2. *Let w be lattice-compatible and suppose each h_ℓ, $\ell \in A$, is absolutely continuous. Then, whatever the dependence of π on income,*

$$|q_k - P_k| \;\le\; \sum_{\ell \in A:\ w/\ell \text{ even}} \frac{\ell^2}{8}\Big(\sup_{|x - b_k^-| \le \ell/2} |h_\ell'(x)| + \sup_{|x - b_k^+| \le \ell/2} |h_\ell'(x)| \Big),$$

where $b_k^{\pm} = kw \pm w/2$ are the boundaries of C_k. In particular $q_k = P_k$ when w/ℓ is odd for every active level.

Proof. Reports at level ℓ never leave the cell of the latent amount except at a boundary b that is a multiple of ℓ. Latent amounts in $[b - \ell/2, b + \ell/2)$ are then reported at b, and half of that mass is assigned to each side. The error at b for the cell on its left is therefore $\frac{1}{2}\int_0^{\ell/2} \{h_\ell(b+t) - h_\ell(b-t)\}\, dt$, whose absolute value is at most $\frac{1}{2} \sup |h_\ell'| \int_0^{\ell/2} 2t\, dt = \frac{\ell^2}{8} \sup |h_\ell'|$. Each cell has two boundaries. □

Without the splitting convention the error is of first order, $\frac{1}{2}\ell\, h_\ell(b)$ at each boundary, and at $w_0 = \$50{,}000$ with an active level of $\$25{,}000$ it is a sizeable fraction of a cell. With the convention it is of second order in the level and vanishes wherever the density is flat. Levels that are in use but excluded from A because their probability is below δ contribute a first-order error proportional to that probability; this is what the cut-off δ of Section 2 has to control.

The least common multiple $w_0 = \operatorname{lcm} A$ is the smallest lattice-compatible width. It is a statement about the data and not about any test: it is the resolution available to an analyst who is unwilling to model how the level depends on income. A width that is lattice-compatible cannot be replaced by a merely coarse one. If reports pile up at multiples of $\$10{,}000$ and of $\$25{,}000$, cells of width $\$10{,}000$ contain two multiples of $\$25{,}000$ in some cases and none in others, and the cell vector inherits a saw-tooth that has nothing to do with g. Any multiple of a lattice-compatible width is again lattice-compatible, so the errors in

choosing the width are not symmetric: a width that is too coarse costs resolution, and one that is too fine reintroduces the saw-tooth.

3.2 Binning preserves unimodality

The next lemma turns a statement about the latent density into one about a probability vector. It is the two-cell case of the variation-diminishing property of the indicator kernel (Karlin, 1968); we give a direct proof.

Lemma 3. *Let g be unimodal with mode M, that is, non-decreasing on $(-\infty, M]$ and non-increasing on $[M, \infty)$, and let $w > 0$. Then (P_k) is a unimodal sequence for every width w and every position of the grid.*

Proof. Let m be the index with $M \in C_m$ and $t = (mw + w/2) - M \in (0, w]$. For $k \le m-2$ the cells C_k and $C_{k+1} = C_k + w$ lie to the left of M, where g is non-decreasing, so $P_k \le P_{k+1}$; symmetrically $P_k \ge P_{k+1}$ for $k \ge m+1$. It remains to show that $P_m \ge \min(P_{m-1}, P_{m+1})$. The interval $[M-w, M]$ is the translate of $C_{m-1} = [M+t-2w, M+t-w)$ by $w - t \ge 0$ to the right, and g is non-decreasing to the left of M; hence $\int_{M-w}^{M} g \ge P_{m-1}$, and since the average of a non-decreasing function over the right-hand part of an interval is at least its average over the whole,

$$\int_{M-(w-t)}^{M} g \;\ge\; \frac{w-t}{w} \int_{M-w}^{M} g \;\ge\; \frac{w-t}{w}\, P_{m-1}.$$

Symmetrically $\int_{M}^{M+t} g \ge (t/w) P_{m+1}$. Adding, $P_m \ge \frac{w-t}{w} P_{m-1} + \frac{t}{w} P_{m+1} \ge \min(P_{m-1}, P_{m+1})$. □

Together, Lemmas 2 and 3 say that at a lattice-compatible width a unimodal latent density produces a reported cell vector that is unimodal up to the second-order boundary term. The converse fails: a unimodal cell vector is consistent with a latent density that has a valley inside a single cell. A test at width w therefore answers the question whether the

latent distribution has a departure from unimodality that is visible at resolution w, and no more.

3.3 Where a valley is, and the Lorenz curve

When the cell vector is not unimodal, let $j_1 < i_2$ be the last cell of the first modal block and the first cell of the second (Section 4), so that some cell m between them has $P_m < \min(P_{j_1}, P_{i_2})$. A density whose average over C_{j_1} exceeds its average over C_m takes at some point $a \in C_{j_1}$ a value larger than it takes at some point $b \in C_m$, and likewise for a point $c \in C_{i_2}$; the density therefore has an essential local minimum in (a, c). Every latent density consistent with the cell probabilities thus has a local minimum in $(c_{j_1} - w/2,\, c_{i_2} + w/2)$, where $c_k = kw$, and the population share below it lies in

$$p^* \in \Big[\textstyle\sum_{k<j_1} P_k,\ \sum_{k\le i_2} P_k \Big]. \tag{4}$$

This interval does not shrink as n grows: it is an outer bound on the identified set, and its width measures what the rounding costs. For readers who work with Lorenz curves rather than densities we note that $L''(p) = 1/\{\mu f(Q(p))\}$, where L is the Lorenz curve, μ the mean and Q the quantile function (Parzen, 1979); a local minimum of the density is therefore a local maximum of the curvature of the Lorenz curve, and (4) locates it on the percentile scale.

4 Testing unimodality at a certified width

4.1 The statistic

Let $p = (p_1, \ldots, p_K)$ be a vector of cell probabilities and let $\mathcal{U}$ be the set of unimodal sequences, those for which $p_1 \le \cdots \le p_m \ge \cdots \ge p_K$ for some m. For $\lambda \ge 0$ and $k \in \{1, 2\}$

let

$$E_k(p,\lambda) = \max_{C} \sum_{i \in C} (p_i - \lambda),$$

the maximum running over unions of at most k sets of consecutive indices, and define the discrete excess mass difference

$$\Delta(p) = \sup_{\lambda \geq 0} \big\{ E_2(p,\lambda) - E_1(p,\lambda) \big\},$$

the analogue for a cell vector of the functional of Müller and Sawitzki (1991). Every upper level set of a unimodal sequence is a set of consecutive indices, so $\Delta(p) = 0$ for $p \in \mathcal{U}$; if $p \notin \mathcal{U}$ there are $i < j < k$ with $p_j < \min(p_i, p_k)$, and any λ between p_j and $\min(p_i, p_k)$ gives $\Delta(p) > 0$. Thus $\Delta(p) = 0$ if and only if $p \in \mathcal{U}$. For fixed λ both maxima are computed in $O(K)$ operations from running extrema of partial sums, and the maximizing pair of blocks delivers the indices $j_1 < i_2$ of (4). The supremum is attained on the finite set of levels at which two candidate blocks have equal excess mass (Lemma S1 of the supplement), and we compute it exactly by following the breakpoints of the piecewise linear functions E_1 and E_2.

Given reports, let $\hat{p}$ be the weighted cell proportions at the certified width, with boundary reports split as in Lemma 2, over the bulk of the distribution: cells are retained from the one containing the first percentile to the one containing the 95th, so that a handful of observations in a sparse tail cannot create a valley. The null hypothesis is $H_0 : p \in \mathcal{U}$, which by Lemmas 2 and 3 is implied, up to the boundary term, by unimodality of the latent density. When fewer than four cells remain the question is not posed and the procedure reports that the resolution is insufficient.

4.2 Calibration on the boundary of the null

The difficulty is that $\mathcal{U}$ is a union of polyhedral cones with a boundary, and the distribution of $\Delta(\hat{p})$ depends on where in $\mathcal{U}$ the truth lies. If p is strictly unimodal, small perturbations remain unimodal and $\Delta(\hat{p}) = 0$ with probability tending to one. If p has a flat stretch,

perturbations of order $n^{-1/2}$ create valleys inside it, and $\sqrt{n}\,\Delta(\hat{p})$ has a nondegenerate limit. Precisely, call a pair of adjacent cells $(k, k+1)$ a *contact* of $p \in \mathcal{U}$ if $p_k = p_{k+1}$, unless these two cells form the entire top plateau of p (that is, $p_{k-1} < p_k = p_{k+1} > p_{k+2}$, with $p_0 = p_{K+1} = -\infty$). A flat stretch whose neighbour is higher on one side, or which is at least three cells long, lets a perturbation dig a valley; a top plateau of exactly two cells does not. Let $\mathcal{T}(p)$ be the set of contacts.

Resampling from $\hat{p}$ fails, because $\hat{p}$ is outside $\mathcal{U}$ with non-vanishing probability (Andrews, 2000). Resampling from the projection of $\hat{p}$ on $\mathcal{U}$ fails more subtly: within a flat stretch the projection is itself random, retaining some ties and breaking others, and the bootstrap then reproduces the behaviour of the statistic at a random nearby point of $\mathcal{U}$ rather than at p. We therefore first decide which adjacent cells are tied. Let $\kappa_n \to \infty$ with $\kappa_n/\sqrt{n} \to 0$; pool adjacent cells k and $k+1$ whenever

$$|\hat{p}_{k+1} - \hat{p}_k| \le \kappa_n \{(\hat{p}_k + \hat{p}_{k+1})/n\}^{1/2},$$

replacing each maximal run of pooled cells by its average, and let $\bar{p}_n$ be the least-squares projection of the result on $\mathcal{U}$. This is generalized moment selection (Andrews and Soares, 2010) applied to the adjacent differences; we use $\kappa_n = (\log n)^{1/2}$. The bootstrap draws $\hat{p}^* \sim \mathrm{Mult}(n, \bar{p}_n)/n$, and the test rejects when $\Delta(\hat{p})$ exceeds the $(1-\alpha)$ quantile c_n^* of $\Delta(\hat{p}^*)$ given the data. The projection is computed by pool-adjacent-violators on either side of each candidate mode, in $O(K^2)$ operations.

Theorem 2. *Let K be fixed and $\hat{p}$ the vector of cell frequencies of n independent observations with cell probabilities p. Let $Z \sim N(0, \mathrm{diag}(p) - pp^\top)$.*

(a) For every $p \in \mathcal{U}$ there are $r > 0$ and a continuous, positively homogeneous function Δ'_p such that $\Delta(p+x) = \Delta'_p(x)$ whenever $\|x\| < r$ and $p + x$ is in the simplex. The function Δ'_p depends on p only through the signs of the adjacent differences $p_{k+1} - p_k$, and $\Delta'_p \equiv 0$ if and only if $\mathcal{T}(p) = \emptyset$.

(b) If $p \in \mathcal{U}$ and $\mathcal{T}(p) = \emptyset$, the probability of rejection tends to zero.

(c) If $p \in \mathcal{U}$, $\min_k p_k > 0$ and $\mathcal{T}(p) \neq \emptyset$, then $\sqrt{n}\,\Delta(\hat{p}) \xrightarrow{d} \Delta'_p(Z)$, and the conditional law of $\sqrt{n}\,\Delta(\hat{p}^*)$ converges in probability to the same limit. Consequently $\lim_n \mathbb{P}(\textit{reject}) = \min\{\alpha, \mathbb{P}(\Delta'_p(Z) > 0)\} \leq \alpha$.

(d) If $p \notin \mathcal{U}$, the probability of rejection tends to one.

The proof, in the supplement, uses that Δ is piecewise linear on the simplex. Near a point of $\mathcal{U}$ it is therefore exactly conic, and its local form depends only on which adjacent cells are tied: a valley can only be dug inside a flat stretch, and the size of the valley depends only on the perturbation restricted to that stretch and on whether each end of the stretch is closed by a lower neighbour or open to a higher one. Moment selection recovers the sign pattern of the adjacent differences of p with probability tending to one, and $\bar{p}_n - p$ is then constant on each flat stretch, which leaves the local form unchanged; the bootstrap is therefore drawn from a point with the same local geometry as p, and the two limits coincide. The limit law of $\Delta'_p(Z)$ has an atom at zero, which makes the test conservative when the atom exceeds $1 - \alpha$.

Four remarks delimit the result. First, it is pointwise in p. Sequences of cell vectors whose adjacent differences shrink at the rate $n^{-1/2}$ are treated as tied, and we have no proof that this is conservative uniformly over $\mathcal{U}$; the simulations of Section 6 include exactly flat designs and designs with shallow valleys to show what happens there. Second, it is stated for a fixed number of cells. When the certified width is fine, as for monthly housing costs or for disposable income, K runs to hundreds, adjacent differences are small relative to their standard errors, and moment selection pools long stretches; the test is then conservative, and Section 6 measures its size and power at $K \approx 1000$. Third, the retained range of cells is set by sample percentiles; it equals its population counterpart with probability tending to one when the population percentiles fall in the interior of cells, and we treat it as fixed. Fourth, the bootstrap draws multinomial samples of size n_{eff}. This matches the first-order behaviour of

weighted cell proportions when the design effect is homogeneous across cells; under a complex design a replicate-weight bootstrap should replace it, with the same moment-selected centre.

Finally, the width is estimated from the same sample. With probability tending to one $\hat{w}_0$ lies in the finite set of lattice-compatible multiples of w_0 (Section 2), and Theorem 2 applies at each of them. It does not cover the data-dependent choice among them; the simulations of Section 6 include that choice.

5 Resolution finer than the certified width

5.1 The reporting map and its inverse

Below the certified width the reported cell probabilities are a linear image of the latent ones through a map that depends on the profile $\pi_\ell(\cdot)$. Fix a width w that divides every active level, take cells centred at the multiples of w, and treat the density as constant within a cell. A latent amount in the cell centred at c is reported, at level ℓ, at the multiple of ℓ nearest to it, which is c itself when $\ell \mid c$, or, if the cell straddles a rounding boundary, at the two nearest multiples in equal parts. Writing P and q for the latent and reported cell vectors,

$$q = A(w)P, \qquad A(w)_{vc} = \sum_{\ell \in A} \pi_\ell(c)\, a_\ell(v, c), \tag{5}$$

where $a_\ell(v, c) \in \{0, \frac{1}{2}, 1\}$ is the share of cell c that level ℓ reports into cell v. For levels that divide w we set $a_\ell(v, c) = \mathbf{1}\{v = c\}$, neglecting the second-order boundary term of Lemma 2. For c a multiple of w let $\rho(c) = \max\{\ell \in A : \ell \mid c\}$ be its roundness.

Theorem 3. *Suppose the active ladder A is totally ordered by divisibility and $w \in A \cup \{1\}$. Then, after ordering the cells by increasing roundness, $A(w)$ is block lower triangular with diagonal diagonal blocks, and*

$$\det A(w) = \prod_{c} \sum_{\ell \in A:\ \ell \mid c} \pi_\ell(c).$$

In particular $A(w)$ is invertible, and the latent cell vector at width w is identified from the reports and the profile, whenever every cell has positive probability of being reported at a level that divides its centre; it suffices that $\sum_{\ell \le w} \pi_\ell(c) > 0$ for every c.

Proof. Let c be a cell centre and $\ell \in A$. If $\ell \mid w$, level ℓ contributes $\pi_\ell(c)$ to the diagonal by the convention above. If $\ell \nmid w$ then $\ell > w$, because A is a chain containing w or $w = 1$; if moreover $\ell \mid c$, the rounding interval of level ℓ around c contains the cell, so every amount in it is reported at c and again contributes $\pi_\ell(c)$ to the diagonal. If $\ell \nmid c$ then, A being a chain, no level at least as coarse as ℓ divides c, so $\rho(c) < \ell$; the cells that receive this mass are centred at multiples of ℓ, whose roundness is at least $\ell > \rho(c)$. Every off-diagonal entry of column c therefore lies in a row of strictly greater roundness, which gives the triangular structure, and entries between distinct cells of equal roundness vanish. The determinant of a block triangular matrix with diagonal blocks is the product of its diagonal entries. □

The identified width is thus the finest level in use, not the least common multiple: for a ladder whose finest non-trivial level is \$100 that is a factor of five hundred below $w_0 =$ \$50,000. The finest level is also the limit, since densities that differ only within cells of width ℓ are observationally equivalent when no respondent reports more finely than ℓ. For ladders that are not chains, such as those containing both \$10 and \$25, the ordering argument fails (rounding \$75 to the nearest \$10 produces \$80, which is less round than \$75), and invertibility is verified numerically. The condition number $\kappa(A(w))$, computable from the estimated mechanism before any data are inverted, prices the resolution: for the ladder estimated from California wages in the ACS (Section 7), with its mixture weights used as a constant profile over incomes up to \$250,000, it is 1.1 at $w =$ \$25,000, 1.6 at \$5,000, 5.5 at \$1,000, 38 at \$500 and 190 at \$100.

The inverted histogram is $\hat{P} = \arg\min\{\|\hat{A}(w)P - \hat{q}\|^2 : \ P \ge 0, \ \mathbf{1}^\top P = 1\}$, computed by projected gradient, and the test statistic is the excess mass difference of $\hat{P}$ over its bulk. Sampling error lives on the report scale, so the bootstrap draws $q^* \sim \mathrm{Mult}\{n_{\mathrm{eff}}, \hat{A}(w)\tilde{P}\}/n_{\mathrm{eff}}$ from the image of the unimodal projection $\tilde{P}$ of $\hat{P}$ and inverts each draw; the variance that

the inversion adds is thereby carried by the null distribution.

5.2 Which profile the inversion needs

The map (5) involves $\pi_\ell(c) = \mathbb{P}(L = \ell \mid X \in C_c)$, the profile conditional on latent income. An estimator that conditions on the report returns a different function, since $\mathbb{P}(L = \ell \mid Y = y) = 0$ whenever $\ell \nmid y$: under the mechanism of our simulations, at \$12,000 the latent-conditional profile over the levels $\{1, 100, 1000, 5000, 10^4\}$ is $(0.153, 0.136, 0.339, 0.169, 0.203)$, while the report-conditional one is $(0, 0.038, 0.961, 0, 0)$. The right object is nevertheless identified: fitting the mixture of Proposition 1 on bands of reports whose edges are multiples of W estimates band averages of the latent-conditional profile, up to the mass that rounding carries across the band edges (Proposition S1 of the supplement).

Identification is not the binding constraint; precision is. The inverted histogram amplifies an error in the profile into an alternation of the cell probabilities with the period of the coarse levels, which the bootstrap, treating the profile as known, does not account for; in Section 6 the inverted test holds its level at $n = 200{,}000$ when handed the true latent-conditional profile, while with an estimated one—a logit fitted to the reports, the band estimator of Proposition S1(b), or the constant marginal weights—its rejection rate on some unimodal design grows with n, a fixed error of a few percentage points in the profile becoming detectable. The inversion is therefore a way of using an external estimate of the profile, from a validation or reinterview study that observes both the latent amount and the report, rather than a procedure that can be run from the reports alone at large sample sizes.

The inverted test also fails in the confident direction when the ladder it is given is wrong: omitting the \$5,000 level, with the true profile of the remaining levels, drives its rejection rate to between 0.95 and one on four of the five unimodal designs of Section 6, while the test at the certified width, which estimates the active set rather than assuming it, is unaffected. Theorem 1 supplies the gate: we invert only when the specification test does not reject on the active ladder; it rejects for every American variable of Section 7.

6 Simulations

6.1 Design

Latent incomes are drawn from seven unimodal and eleven bimodal distributions. The unimodal ones are a lognormal (LN, median \$40,135), a gamma (GA), a Singh–Maddala (SM), a lognormal mixture with a shoulder on its upper side (SHLD), a density with a nearly flat stretch between \$50,000 and \$95,000 (PLAT), and two piecewise linear densities that are exactly flat, one on the decreasing side of the mode (FLATDEC) and one at the top (FLATTOP). The last two put the cell vector on the boundary of $\mathcal{U}$, where Theorem 2(c) applies. The bimodal distributions are mixtures of two lognormals with valleys \$6,000 to \$16,000 wide (N1–N3), \$25,000 to \$108,000 wide (M1–M6, including an unequal mixture and a low-income cluster), or \$52,000 wide but shallow, the density at the valley floor being 0.85 and 0.66 of the lower peak (S1, S2).

Four reporting mechanisms are used. *Exact*: no rounding. *ACS*: the ladder $\{1, 100, 1000, 5000, 10^4\}$ with a profile that moves weight towards coarse levels as income rises, calibrated so that 84, 43 and 28 per cent of reports are multiples of \$1,000, \$5,000 and \$10,000; $w_0 =$ \$10,000 and the residues of every active level are close to uniform. *ACS + \$25k*: the same, with a quarter of respondents rounding to \$25,000, so that $w_0 =$ \$50,000 and between a quarter and a half of the reports lie below W. *Types*: a third of respondents round coarsely whatever their income and the rest almost never, and the ladder contains \$2,500. The sample sizes are 2,000 and 10,000, with 400 replications for unimodal and 200 for bimodal designs, at nominal level five per cent. The candidate ladder is $\{1, 10, 100, 500, 1000, 2500, 5000, 10^4, 2.5 \cdot 10^4\}$ and δ_n is as in Section 2.

Besides the cell test of Section 4 we report five procedures: the excess mass test calibrated as in Ameijeiras-Alonso et al. (2019) and the dip test calibrated as in Cheng and Hall (1998), both on the logarithm of the reports; and three natural repairs, the critical-bandwidth test with the bandwidth bounded below by 0.7 times the coarsest active level, at which a Gaussian

Table 1: Rejection frequencies at the five per cent level, $n = 10{,}000$; 400 replications for unimodal designs (upper panel), 200 for bimodal designs (lower panel). EM: excess mass; dip: dip test; floor: critical bandwidth with a bandwidth floor; jit.: excess mass after jittering; rer.: excess mass calibrated against a re-rounded null; cell: the test of Section 4 at $\hat{w}_0$, counted as not rejecting when the procedure declines. Monte Carlo standard errors are about 0.011 and 0.015 at a rate of 0.05. †Unimodal in levels but not in logarithms, the scale on which the first five procedures are applied.

	ACS mechanism						ACS + \$25k					
design	EM	dip	floor	jit.	rer.	cell	EM	dip	floor	jit.	rer.	cell
LN	1.00	1.00	.24	.05	.01	.00	1.00	1.00	.01	.30	.19	.00
GA	1.00	1.00	.41	.05	.01	.00	1.00	1.00	.11	.06	.76	.00
SM	1.00	1.00	.20	.03	.27	.00	1.00	1.00	.18	1.00	.01	.00
SHLD†	1.00	1.00	.24	.21	.01	.00	1.00	1.00	.00	.47	.01	.00
PLAT	1.00	1.00	.18	.04	.00	.00	1.00	1.00	.00	.06	.55	.00
FLATDEC†	1.00	1.00	.29	1.00	.00	.01	1.00	1.00	.32	1.00	.00	.00
FLATTOP	1.00	1.00	.27	.02	.00	.03	1.00	1.00	.26	.02	.03	.00
N1	1.00	1.00	1.00	1.00	1.00	.00	1.00	1.00	.14	1.00	1.00	.00
N2	1.00	1.00	.99	1.00	1.00	.00	1.00	1.00	.00	1.00	1.00	.00
N3	1.00	1.00	1.00	1.00	1.00	.00	1.00	1.00	1.00	1.00	1.00	.00
M1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.00
M2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.00
M3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.00
M4	1.00	1.00	1.00	1.00	.93	1.00	1.00	1.00	.00	1.00	.38	.00
M5	1.00	1.00	1.00	1.00	.00	1.00	1.00	1.00	.00	1.00	.02	.00
M6	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.97	1.00
S1	1.00	1.00	1.00	1.00	.62	.74	1.00	1.00	.00	1.00	.95	.00
S2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.00	1.00	.43	.00

kernel removes a lattice of that spacing; the excess mass test applied after each report is replaced by a uniform draw over the rounding interval implied by a level drawn from its posterior given the report's pattern; and the excess mass statistic of the reports calibrated against a unimodal null passed through the estimated mechanism.

6.2 Results

Table 1 reports the rejection frequencies at $n = 10{,}000$ under the two ACS mechanisms; Tables S3–S5 of the supplement cover exact and type-dependent reporting and $n = 2{,}000$.

Under all three rounding mechanisms the excess mass and dip tests reject unimodality in every one of the 16,800 replications drawn from the seven unimodal designs. With exact

reports they hold their level on the four designs that are unimodal on the logarithmic scale, with rejection rates between 0.01 and 0.06 (the dip test is oversized on the plateau, 0.30–0.35), and correctly reject on SHLD and FLATDEC, which are unimodal in levels but not in logarithms.

The repairs help only when the mechanism is benign. A bandwidth floor removes the heaps and much of the signal with them, and its size ranges from 0.18 to 0.41 under the ACS mechanism. Jittering over the posterior rounding interval holds its level on the designs unimodal in logarithms under the ACS and type mechanisms, but not once a quarter of respondents round to \$25,000, where it rejects in 30 per cent of lognormal and every Singh–Maddala sample. Calibrating the usual statistic against a re-rounded null is the strongest competitor: under the ACS mechanism it has power close to one against every bimodal design except the low-income cluster, including the narrow valleys the cell test cannot see, but its level rests on the estimated mechanism being right, reaching 0.27 on the Singh–Maddala design under the ACS mechanism and 0.55 and 0.76 on the plateau and gamma designs when the \$25,000 level is added.

The cell test holds its level everywhere. Across the seven unimodal designs, four mechanisms and two sample sizes its largest rejection rate is 0.070, on FLATDEC with exact reports at $n = 2{,}000$, a design on the boundary of the null; the next largest is 0.048. Resampling from the unmodified projection instead of the moment-selected centre raises the rate on the same design to 0.12 and 0.09, the failure Theorem 2 predicts. When the \$25,000 level is active the certified width is \$50,000, three or four cells remain, and the procedure declines or accepts: it cannot be wrong in the confident direction.

The cell test pays for this in resolution. At $\hat{w}_0 = \$10{,}000$ it has power one against valleys \$25,000 wide or wider and against the deeper shallow valley, and 0.74 against the valley whose floor is 85 per cent of the lower peak. Valleys narrower than the grid (N1–N3) are not identified at that width; there the fitted ladder contains a level too coarse for the support and the procedure reports that no lattice-compatible grid exists. At \$50,000 only the widest

valley (M6) is detected. With exact reports, where $\hat{w}_0 = 1$ and the test works on five hundred to a thousand cells, its power is one against every bimodal design but S1 (0.65).

The specification test's rejection rate on correctly specified mechanisms lies between 0.03 and 0.09 in every configuration of Table S1; it rejects a ladder that omits the active levels \$500, \$2,500 and \$5,000 with probability one at $n = 2{,}000$, and under the \$25,000 mechanism its rejection rate increases with the share of reports below W: 0.05 when that share is two per cent, 0.30 at 23 per cent and $n = 20{,}000$, one when it exceeds a third. The activity cut-off is calibrated in Table S2: ignoring a \$25,000 level used by two per cent of respondents leaves the cell test at its level up to $n = 10{,}000$, and at $n = 50{,}000$ a one per cent level is the most that can be ignored, which is the origin of δ_n.

The inversion of Section 5 was run at $w = \$5{,}000$ under a profile that moves strongly with income, at $n = 50{,}000$ and 200,000 (Table S6; 60 replications, so these rates are indicative). With the true profile it holds its level, with rejection rates of at most 0.03 on the unimodal designs; with the logit profile the rate on the plateau rises from 0.18 to 0.40 between the two sample sizes, with the band profiles it lies between 0.30 and one, and with the marginal profile it is one. Omitting the \$5,000 level from an otherwise correct profile drives it to between 0.95 and one on four of the five designs on which this was tried.

7 Empirical applications

7.1 Data and settings

All data are public-use files of the U.S. Census Bureau. From the 2022 one-year ACS Public Use Microdata Sample we take persons aged 25 to 64 in California, Mississippi and Puerto Rico, with annual wage and salary income (WAGP) and total personal income (PINCP), and occupied housing units in California, with monthly owner costs (SMOCP), contract rent (RNTP) and gross rent (GRNTP). From the 2023 CPS Annual Social and Economic Supplement we take wage and salary income of persons aged 25 to 64, and from the ASEC

files of 2017 to 2021 (income years 2016 to 2020) one-person households living in housing units, with disposable income defined as total money income less federal income tax after credits, state income tax and payroll tax. All computations use the survey weights, with the effective sample size of Section 2 in the specification test and in the bootstrap.

The candidate ladder is $\mathcal{L} = \{1, 2, 5, 10, 25, 50, 100, 250, 500, 1000, 2500, 5000, 10^4, 2.5{\cdot}10^4, 5{\cdot}10^4\}$, the cut-off is $\delta_n = \min\{0.02, 2n_{\text{eff}}^{-1/2}\}$, and the specification test uses $M = 10$ arcs and every candidate level that divides the working width, the least multiple of $\hat{w}_0$ of the form $\{1, 2, 5\} \times 10^k$ that leaves at most a thousand cells between the first and the 95th percentile; it differs from $\hat{w}_0$ only when no level is active or the active levels are fine. Specification p-values use 499 conditional draws, so the smallest attainable value is 0.002; the cell test uses 499 bootstrap draws (999 for the CPS one-person households).

The Census Bureau rounds income amounts in the PUMS before release: amounts between \$8 and \$999 to the nearest \$10, amounts up to \$49,999 to the nearest \$100, and larger amounts to the nearest \$1,000 (U.S. Census Bureau, 2024); in the 2022 files every positive WAGP below \$50,000 and above \$999 is a multiple of \$100 and every one from \$50,000 upwards a multiple of \$1,000. In the model of Section 2 this is an admissible level that depends deterministically on income, but it assigns the level \$100 only to amounts below \$50,000, the least common multiple of the ladder: the multipliers of that level lie in a single period of $W/100$, and Assumption 1 fails for it by construction. PINCP is a sum of rounded components and inherits a weaker version of the same structure. The CPS public-use file applies no such rule; seven per cent of its positive wage reports are not multiples of \$100.

7.2 Earnings and personal income

The excess mass and dip tests applied to the logarithm of the reports, on a weighted resample of 6,000, return p-values of zero for every variable in Table 2. In California, Mississippi and the CPS the \$25,000 or \$50,000 level is active and $\hat{w}_0 = \$50{,}000$; in Puerto Rico $\hat{w}_0 = \$10{,}000$. The specification test rejects on every variable: where $\hat{w}_0 = \$50{,}000$, between 43 and 67 per

Table 2: Earnings and personal income. Shares are percentages of positive reports divisible by \$1,000, \$5,000 and \$10,000; "below" is the share of reports below $\hat{w}_0$; T/df and d are the specification statistic divided by its degrees of freedom and divided by the effective sample size; K is the number of cells at $\hat{w}_0$ and p the p-value of the cell test. All specification p-values are 0.002, the smallest attainable.

				% divisible by					specification			cell test	
source	variable	n	median	1k	5k	10k	$\hat{w}_0$	below	T/df	df	d	K	p
ACS CA	wages	148,347	52,000	88	55	38	50,000	0.46	94.90	60	0.060	6	1.00
ACS CA	pers. income	178,286	50,000	76	45	30	50,000	0.50	91.00	69	0.056	6	1.00
ACS MS	wages	9,632	40,000	85	44	29	50,000	0.63	8.70	51	0.083	4	1.00
ACS MS	pers. income	12,389	35,400	70	35	22	50,000	0.67	14.30	60	0.126	4	1.00
ACS PR	wages	7,527	21,400	46	18	12	10,000	0.16	11.80	33	0.071	10	1.00
ACS PR	pers. income	11,148	18,300	39	15	9	10,000	0.24	21.00	33	0.085	9	1.00
CPS 2023	wages	55,499	55,000	86	54	35	50,000	0.43	8.30	105	0.022	5	1.00

cent of the reports lie below it, so the multipliers of the coarse levels occupy the first one or two periods of W and cannot be close to uniform within their classes; in the ACS the agency's rounding adds a failure by construction, and the distances d are between two and a half and six times that of the CPS. In Puerto Rico only a sixth of wage reports lie below $\hat{w}_0 = \$10{,}000$, and the rejection has an institutional source: the PUMS replaces amounts above a state's top-coding threshold by their mean (U.S. Census Bureau, 2024), which places 1.6 per cent of the weighted wage reports at \$241,000, a single residue modulo \$10,000, the case of mass that is not the result of rounding.

With four to ten cells at the certified width the cell test finds nothing; this is not evidence of unimodality but the statement that the reports carry no information about modality at a resolution they support without a model of the profile. Excluding allocated reports changes none of these conclusions, except that for personal income in Mississippi the \$50,000 level falls below δ, the certified width becomes \$10,000, and the cell test on its 14 cells returns $p = 0.13$. The inversion of Section 5 would buy resolution, since $\kappa(A(w))$ is modest down to \$1,000, but its gate is shut: the mechanism it would invert is rejected.

Table 3: Monthly housing costs, ACS 2022, California. Columns as in Table 2; "valley" is the interval, in dollars a month, that must contain a local minimum of the latent density, and p^* the identified set (4) for its percentile. All specification p-values are 0.002.

variable	n	active	$\hat{w}_0$	T/df	df	d	K	p	valley	p^*
owner cost (SMOCP)	85,603	10, 25	50	33.75	24	0.015	112	0.85	1,525–1,775	0.38–0.44
contract rent (RNTP)	50,520	10, 50, 100, 500	500	39.49	15	0.018	8	1.00	–	–
gross rent (GRNTP)	50,520	5, 10, 50, 100	100	13.75	33	0.014	39	0.01	250–650	0.01–0.06

7.3 Housing costs

Monthly housing costs are small amounts measured against fine ladders (Table 3): the certified widths are \$50, \$100 and \$500 a month, and two of the three variables leave the test dozens of cells. The specification test again rejects, but the distances d are 0.014 to 0.018, below even the value for CPS wages: with tens of thousands of reports a departure of this size is detected without being large. Owner costs and gross rent are constructed variables, the first adding mortgage payments, taxes, insurance, utilities and fees, some of them annual amounts divided by twelve (the active \$25 level is the trace of that division), the second adding utilities to contract rent. For owner costs the test does not reject ($p = 0.85$). For gross rent it does ($p = 0.012$), locating the valley between \$250 and \$650 a month, within the bottom six per cent of renters, where a separate concentration of very low gross rents would be consistent with subsidised tenancies. But the mechanism is rejected, so by the rule of Section 2 the finding is kept only if it persists at coarser lattice-compatible widths; at \$200 it does not ($p = 0.38$), and we do not retain it.

7.4 Disposable income of one-person households

Disposable income, the variable on which modality is usually studied, is an amount computed by the survey from reported components, simulated taxes and credits, whose arithmetic owes little to the respondent. Table 4 reports one-person households, for which no equivalence scale intervenes. On the full sample the levels \$10 and \$100 are active and the specification test rejects in every year, with T/df between 7.5 and 13. The lattice enters with the

Table 4: Disposable income of one-person households, CPS ASEC, income years 2016–2020. For each sample: the specification statistic T/df with its p-value, the p-value of the cell test, and the interval (in thousands of dollars) that must contain the valley located by the test. "Filtered" excludes households with more than half of their money income from public transfers.

sample		2016	2017	2018	2019	2020
all	n	16,594	16,320	16,658	14,892	15,701
	spec.	7.52 (< 0.01)	8.94 (< 0.01)	8.09 (< 0.01)	7.85 (< 0.01)	12.95 (< 0.01)
	cell	< 0.01	< 0.01	0.06	0.23	0.56
	valley	17.5–20.4	11.0–13.2	10.9–13.2	27.5–32.1	11.5–12.9
filtered	n	10,698	10,398	10,686	9,657	9,677
	spec.	1.33 (0.11)	1.10 (0.30)	1.40 (0.06)	1.10 (0.31)	1.74 (< 0.01)
	cell	0.01	0.009	0.39	0.003	0.81
	valley	0.1–8.7	0.9–8.7	26.5–30.5	0.5–8.9	2.3–11.3
filtered, gross	n	3,673	3,679	3,876	3,405	3,459
earnings a multiple	spec.	1.53 (0.04)	1.43 (0.06)	1.23 (0.16)	1.76 (< 0.01)	1.93 (< 0.01)
of \$5,000	cell	0.01	< 0.01	< 0.01	0.02	0.03
	valley	25.3–31.1	25.1–30.9	25.7–35.7	25.6–31.9	27.7–33.7
filtered, other	n	7,025	6,719	6,810	6,252	6,218
	spec.	1.57 (0.01)	0.92 (0.60)	1.99 (< 0.01)	1.37 (0.08)	1.13 (0.31)
	cell	0.01	< 0.01	0.27	0.008	0.46
	valley	0.6–9.6	1.3–10.0	2.4–7.2	1.1–10.5	2.3–12.3

households living mainly on public transfers: once they are removed no level is active, and the specification test accepts in four of the five years at the five per cent level. On the full sample the cell test rejects in two years, at valleys whose location moves from year to year; with a rejected mechanism we do not interpret them.

On the filtered sample the test rejects in 2016, 2017 and 2019, each time at a valley below \$9,000, within the bottom six per cent of the distribution; the mechanism passes, but at a width of \$1,000 none of the three rejections survives at the five per cent level ($p = 0.61$, 0.08 and 0.54), and we do not interpret them. A thinning of the distribution between roughly \$25,000 and \$32,000 appears in 2018 on the filtered sample, but is not significant there ($p = 0.39$) and is not what the test finds in the other years. It is significant in every year in the subsample of respondents whose gross earnings are a multiple of \$5,000, and absent from the complementary subsample, where the only valleys the test finds are again at the bottom. The rejection weakens as the width grows past the spacing of the atoms described

next: at $2,000 the test rejects only in 2016, and there between $31,000 and $35,000. A gross amount rounded to $5,000 is carried through the tax computation to an amount that is no longer round: in 2017, gross earnings of $30,000, $35,000 and $40,000 map to disposable incomes with medians $24,400, $28,300 and $31,700, each with an interdecile range of about $1,500. The rounding of the components survives as a sequence of atoms about $3,500 apart, at amounts with no divisibility structure, and at a width of $200 the test sees the spaces between them.

The thinning between $25,000 and $32,000 is thus a property of how gross earnings are reported, not evidence of a gap in the distribution of disposable income. And a specification test applied to a constructed variable cannot see rounding in its components, because the construction destroys the divisibility on which the test rests: on the subsample where the transmitted atoms are strongest, the test accepts the reporting model in two of five years at the five per cent level.

8 Discussion

Rounding is not the only source of atoms. The model treats every pile-up as something the respondent did. A minimum wage times full-time hours, a flat-rate pension, a benefit scale and a top code put genuine mass at particular amounts, and a density with such an atom is not unimodal in any useful sense. The two differ in their arithmetic: rounding places mass at every multiple of a level, an institutional amount at one value or a few unrelated ones. The mixture of Proposition 1 describes the first and the test of Theorem 1 reports when the reports are not explained by it; a procedure estimating both kinds of atom jointly remains to be developed.

The variable matters more than the survey. A sum of amounts reported by different household members, net of simulated taxes and divided by an equivalence scale, retains little of the arithmetic of its components, and what it retains can mislead (Section 7); agencies

that publish constructed totals could release the reported components alongside them.

Limitations. The identification argument assumes that a level is chosen and the amount then rounded to it, which excludes respondents who report a round number unrelated to their income; that the levels lie in a known candidate ladder, protected by including every plausible round amount and by the specification test, which detects an omitted level dividing W; and that residues are close to uniform given the level, which fails when income is narrow relative to W, a failure the specification test detects and the share of reports below W foreshadows. The validity of the cell test is pointwise and for a fixed number of cells; uniform validity over the boundary of the cone, and theory for a number of cells growing with the sample, are open. The inverted test has no asymptotic theory beyond Theorem 3 and should be used only when the specification test passes. Rounding propensity varies with education, pay frequency and proxy response; a statement about such a subpopulation rests on the mechanism within it, and since Proposition 1 conditions only on the level, the estimator extends to covariates directly.

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